FORESIGHT Demand & Inventory Intelligence

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Executive Summary

This project develops a demand forecasting and inventory intelligence solution for NorthBay Living using historical retail sales data. An XGBoost regression model predicts product demand and supports inventory decisions through an interactive Streamlit dashboard.

Problem Statement

Retail businesses face stockouts, overstock, and inefficient inventory planning. Accurate demand forecasting can improve service levels and reduce excess inventory.

Objectives

- Forecast SKU demand
- Identify inventory risk
- Recommend reorder and markdown actions
- Provide an interactive dashboard for decision-making

Dataset

Historical retail transaction data includes product information, quantities sold, revenue, pricing, and dates.

Data Preprocessing

Cleaned data, handled missing values, converted dates, aggregated sales, and prepared data for modelling.

Exploratory Data Analysis

Examined sales trends, product demand distribution, seasonal behaviour, inventory risk, and revenue patterns.

Feature Engineering

Created calendar features (year, month, week, quarter), lag features, rolling averages, and price-related variables for model training.

Model Development

Trained an XGBoost Regressor and compared performance against a Seasonal Naive Forecast baseline.

Model Evaluation

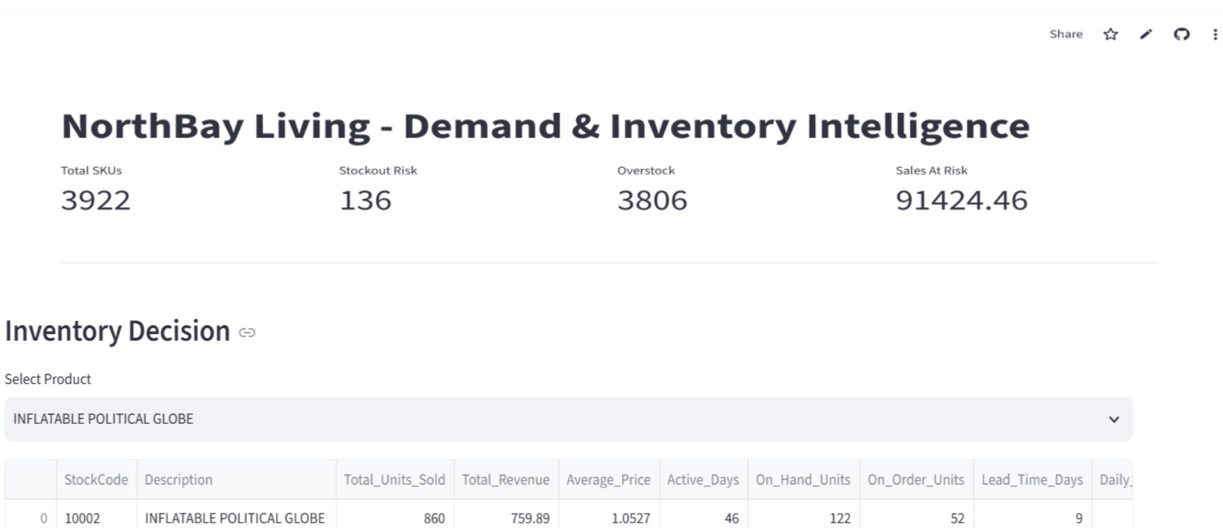
XGBoost: WAPE 20.15%, RMSE 43.72.
Baseline: WAPE 86.48%, RMSE 157.14.
The proposed model substantially outperformed the baseline.

Inventory Intelligence

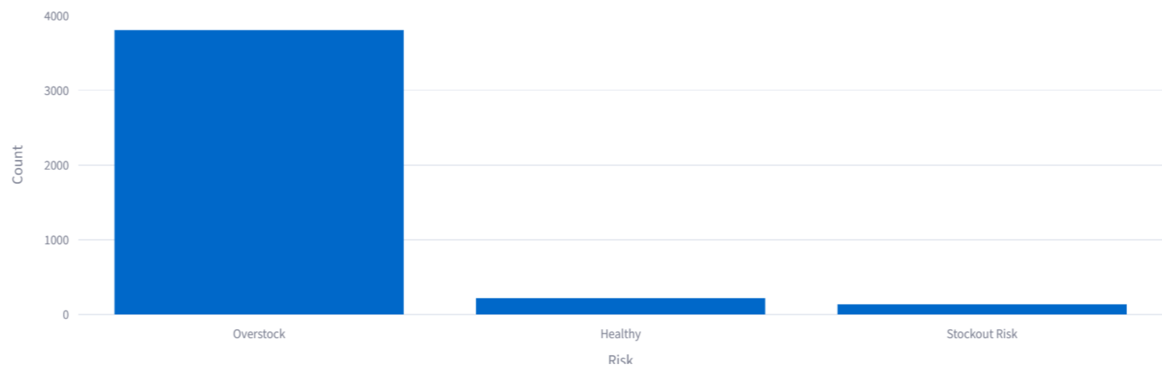
Generated stockout risk, overstock risk, reorder quantity recommendations, sales at risk, and capital locked metrics.

Dashboard

Implemented a Streamlit dashboard displaying KPIs, product selection, inventory decisions, and integrated demand forecasting.



Inventory Risk Distribution

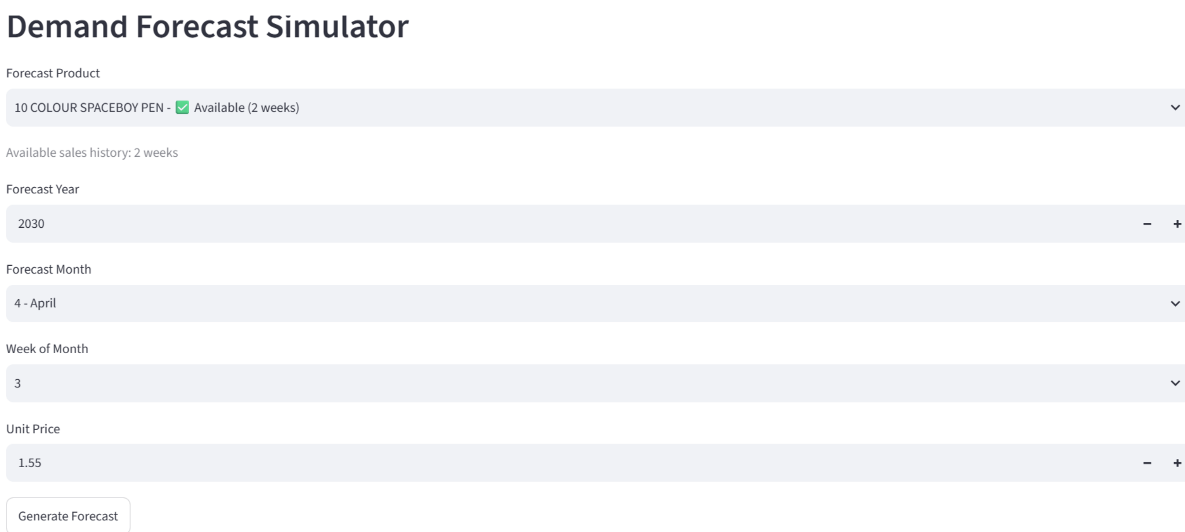


Priority Reorder List

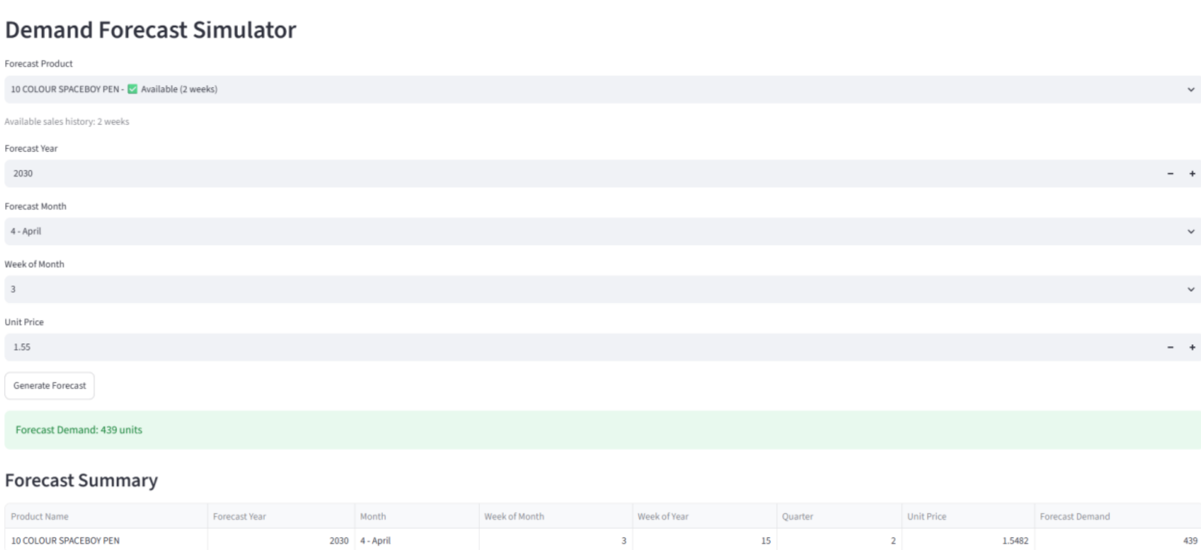
	StockCode	Description	Recommended_Order_Qty
29	16008	SMALL FOLDING SCISSOR(POINTED EDGE)	68
33	16014	SMALL CHINESE STYLE SCISSOR	245
37	16033	MINI HIGHLIGHTER PENS	192
39	16045	POPART WOODEN PENCILS ASST	138
78	16258A	SWIRLY CIRCULAR RUBBERS IN BAG	41
142	20668	DISCO BALL CHRISTMAS DECORATION	996
177	20723	STRAWBERRY CHARLOTTE BAG	200
178	20724	RED RETROSPOT CHARLOTTE BAG	130
179	20725	LUNCH BAG RED RETROSPOT	194
369	21098	CHRISTMAS TOILET ROLL	18

Markdown Clearance List

	StockCode	Description	Capital_Locked
0	10002	INFLATABLE POLITICAL GLOBE	128.4236
1	10080	GROOVY CACTUS INFLATABLE	187.4167
2	10120	DOGGY RUBBER	75.18
3	10123C	HEARTS WRAPPING TAPE	188.5
4	10124A	SPOTS ON RED BOOKCOVER TAPE	52.92
5	10124G	ARMY CAMO BOOKCOVER TAPE	38.22
6	10125	MINI FUNKY DESIGN TAPES	174.5869
7	10133	COLOURING PENCILS BROWN TUBE	27.2606
8	10135	COLOURING PENCILS BROWN TUBE	106.44
9	11001	ASSTD DESIGN RACING CAR PEN	251.1886



Example :-



Product Name	Forecast Year	Month	Week of Month	Week of Year	Quarter	Unit Price	Forecast Demand
10 COLOUR SPACEBOY PEN	2030	4 - April	3	15	2	1.5482	439

Business Impact

Supports better inventory planning, reduces stockouts, minimizes excess inventory, and enables data-driven decisions.

Limitations

Model performance depends on historical data quality and does not explicitly incorporate promotions or external market events.

Future Work

Integrate external variables, automate retraining, add real-time data ingestion, and deploy a production scoring API.

Conclusion

FORESIGHT demonstrates how machine learning and analytics can improve retail demand forecasting and inventory management.

References

GitHub Repository Link : https://github.com/DLS-Nadavi/Zidio-Development_Project-01_foresight-demand-inventory-intelligence.git

Streamlit App Link : https://zidio-developmentproject-01foresight-demand-inventory-intellig.streamlit.app/

Used Dataset - https://www.kaggle.com/datasets/cgrymn/online-retail-ii-uci-dataset?utm_source=chatgpt.com